

Determination of the Rare Earth Elements in Aqueous Samples at Sub-ppt Levels by Inductively Coupled Plasma Mass Spectrometry and Flow Injection ICPMS

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A method has been developed for the determination of the rare earth elements (REE) at concentrations down to 0.1 ppt (0.5 pmol kg⁻¹) in aqueous samples using flow injection coupled to an inductively coupled plasma mass spectrometer. The method presented here is an extension of an existing technique to allow application to a wider range of ionic strengths. The method is sufficiently rapid to enable samples to be processed through the purification stage in less than 90 min. More than 99.5% of the Ba in the sample is removed during processing, ensuring that isobaric interference of BaO on the REE is <2% of the REE signal. The detection limits of the method described here show an improvement of up to 30 times better than the original method. This improvement was achieved by using flow injection techniques that allow the sample to be concentrated in a smaller volume for analysis. The new technique has been successfully employed for the determination of REE in Icelandic hydrothermal fluids.

The ability to detect low levels of rare earth elements (REE) in aqueous samples (e.g., seawater and hydrothermal fluids) is important in the study of the mobility of REE during water–rock interactions. The behavior of the REE may also be used as analogues of the geochemistry of the radioactive actinides in future radioactive waste repositories. Inductively coupled plasma mass spectrometry (ICPMS) is widely used for REE analyses of rocks after they have been brought into solution.¹ However, the detection limits are such that a preconcentration step is required to analyze REE in most natural aqueous solutions. Simple preconcentration of the REE by evaporation of the sample is not generally suitable as it frequently leads to precipitation of salts

(which may also take up REE) and/or it leads to a solution with high total dissolved solids that creates potential isobaric interferences or clogs the nebulizer and sampling cones of the ICPMS with salt deposits. There are many existing methods for the analysis of REE in solution, e.g., ion exchange resins^{2–4} and solvent extraction;⁵ however, many of these techniques do not show complete recovery of the REE or show low enough detection limits.

A method is presented here that allows the determination of REE levels in aqueous samples at concentrations as low as 0.1 ppt (0.5 pmol kg⁻¹). This new method is based on that of Shabani et al.,⁶ but modifications presented here allow the method to be applied to a greater range of natural aqueous solutions, other than the high ionic strength fluids considered in the original technique. Enhanced detection limits, compared to the original method⁶ were achieved by the use of flow injection coupled to an inductively coupled plasma mass spectrometer. This allows the sample to be preconcentrated to smaller fluid volumes than is required for normal nebulization. It can then be injected into the plasma source at precise time intervals. The additional use of an ultrasonic nebulizer can further improve the detection limits; however, this was not available for this work.

EXPERIMENTAL SECTION

When an acidified sample comes in contact with a mixture of alkyl hydrogen phosphates in heptane, the REE are quantitatively transferred from an aqueous phase to an organic phase.⁶ After

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Table 1. Operating Conditions for Determination of REE in Aqueous Samples Using a PlasmaQuad PQ2+ ICPMS and a FIAtron Flow Injector

Plasma Conditions		
forward power		1350 W
reflected power		<5 W
cool gas flow		14 L min ⁻¹
auxiliary gas flow		0.8 L min ⁻¹
nebulizer gas flow		0.75 L min ⁻¹
Sampling Conditions		
standard nickel sample cone		1.2 mm
sample cone aperture		
standard nickel skimmer cone		1.0 mm
skimmer cone aperture		
Nebulizer		
Meinhard nebulizer		0.8 mL min ⁻¹
solution uptake rate		
controlled by peristaltic pump		
Data Acquisition		
mass range		100.9–189.8
dwelt time		160 ms
no. of scans		400
no. of channels		1024
collector type		pulse
Operating Conditions for the Flow Injection		
length of PTFE tubing		906 mm
inside diameter of tubing		0.508 mm
sample loop size		460 μ L
wash time		300 s
load time		30 s
valve to plasma time		34 s
injection time		50 s
sample spread time		51 s

removal of the major elements, the REE are back-extracted into an aqueous phase through the addition of octanol and 6 M HCl to the organic phase. The original method⁶ gives unacceptably low yields of the REE (~70%) when applied to low ionic strength solutions. A number of modifications have been made to increase the yield to more than 98% in low ionic strength aqueous samples such as hydrothermal waters from subaerial hydrothermal systems.

Apparatus. All apparatus used were PTFE and were cleaned by boiling in 50% HNO₃ for 1 h. The PTFE were then rinsed with sub-boiling distilled water (SB water) and left to dry in a laminar flow cabinet. Polypropylene pipet tips (5 mL) were cleaned by boiling in 5% HNO₃ for 1 h and rinsed with deionized–distilled water.

Reagents. The reagents employed in this study were as follows: *n*-heptane (AR grade, Merck); octan-1-ol (99+% HPLC grade Aldrich); diester mix of 65% bis(2-ethylhexyl) hydrogen phosphate and 35% 2-ethylhexyl dihydrogen phosphate (Tokyo Kasei Co. Ltd.); methanol (AR grade, Merck); HNO₃ (AristaR grade, Merck); orthophosphoric acid (AR grade, Merck). HCl (6 M) was produced by a three times distillation of HCl in quartz glassware; 0.25 M HCl was made by dilution of 6 M HCl with SB water. The SB water was manufactured by first boiling deionized water in a quartz still and then sub-boiling distilling, in PTFE under an infrared lamp.

Standards. Calibration of the ICPMS for REE determination was carried out by making working standards from the following stock standards; SpecPure atomic absorption standards (Johnson Matthey) of Lu and Tm at 1000 μ g/mL in 5% HNO₃; atomic absorption standards (Aldrich) of In, Ba, La, Ce, Pr, Nd, Sm, Eu, Gd, Tb, Dy, Ho, Er, Yb, and Re at 1000 μ g/mL in 1% HNO₃; Ru

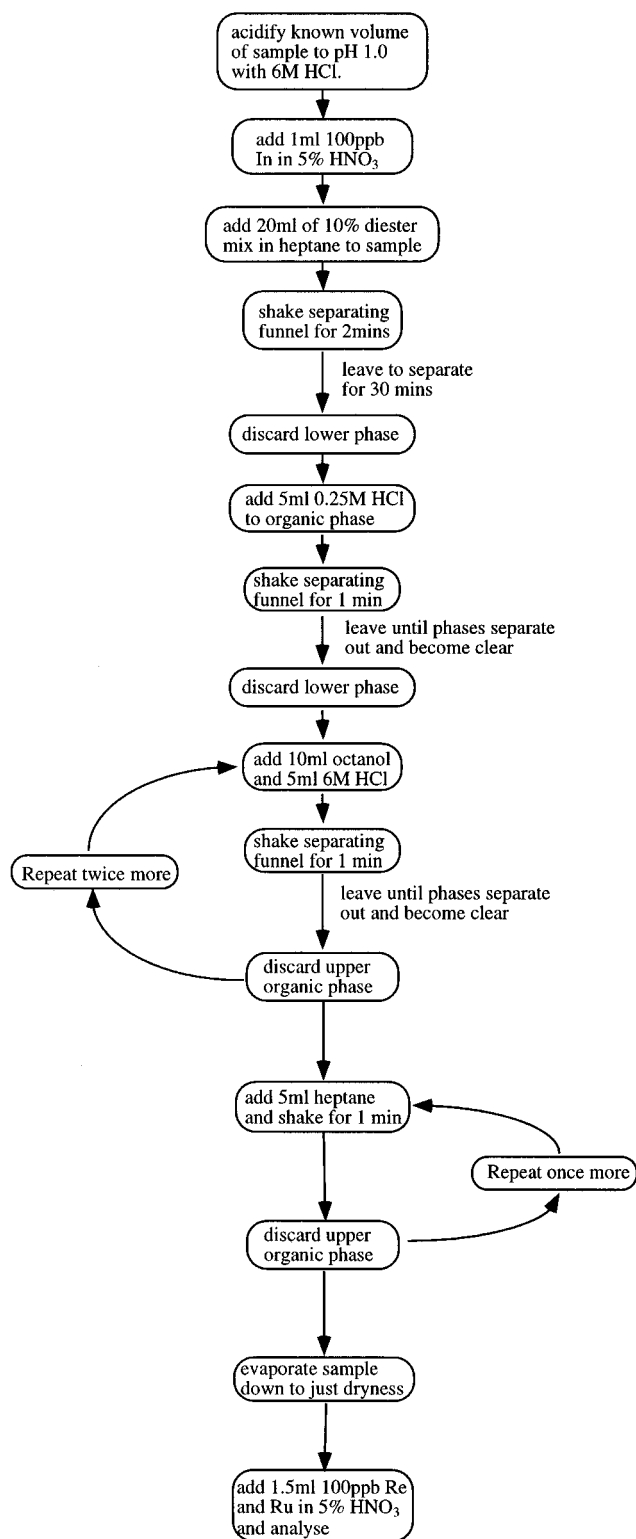


Figure 1. Flow chart indicating sample preparation for the preconcentration technique.

at 1000 μ g/mL in 5% HCl. Standards of REE, Ba, and In were made at the following concentrations in 5% HNO₃: 1, 2, 5, 10, 20, 50, and 100 ppb. All samples contained Re and Ru (internal standards) at 100 ppb concentrations. Standards were made at these concentrations in order to weight the linear regression of the calibration to the low concentrations.⁷ A blank standard with the Re and Ru (internal standards) was also used for calibration.

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ICPMS. The ICPMS used was a PlasmaQuad 2+ (Fisons Instruments), fitted with a concentric Meinhard nebulizer. The operating conditions are shown in Table 1. A Gilson Minipuls 3.0 peristaltic pump was used for sample introduction. The ICPMS was optimized with a solution of ^{115}In at 100 ppb in 5% HNO_3 . The internal standards used during the analysis of samples containing REE were ^{102}Ru , ^{185}Re , and ^{187}Re at 100 ppb concentration in 5% HNO_3 .⁸ The REE and Ba were measured on the following masses: ^{138}Ba , ^{139}La , ^{140}Ce , ^{141}Pr , ^{146}Nd , ^{147}Sm , ^{151}Eu , ^{158}Gd , ^{159}Tb , ^{162}Dy , ^{165}Ho , ^{166}Er , ^{169}Tm , ^{174}Yb , and ^{175}Lu .

Flow Injector. Flow injection was performed by a FIA-tron FIA valve 2000 supplied by Fisons Instruments. The flow injector was fully interfaced to the ICPMS and the operating conditions are shown in Table 1.

Solvent Extraction and Back-Extraction Procedure. A summary of the method is shown as a flow chart in Figure 1. A description of the method is given below and this is modified from that of Shabani et al.⁶

Indium was added to the sample to monitor recovery of the REE² during the extraction process. Sufficient 6 M HCl was added to the sample to reduce the pH to 1.0 ± 0.1 . (In the original method⁶ a pH of 1.5 was suggested, but this was found (see below) to be too high for complete recovery of the REE in low ionic strength solutions.) The original technique⁶ stated that only 2 min was necessary for complete phase separation, and this was found to be inadequate; furthermore, this stage was found (see below) to be an important consideration for complete REE recovery. HCl (0.25 M) was added to the organic phase to remove any remaining major elements into the aqueous phase which was later discarded. After the addition of octanol (10 mL), 6 M HCl (5 mL) was added to back-extract the REE into the aqueous phase. The aqueous phase was collected, and the back-extraction procedure was repeated three more times to ensure complete transfer of the REE into the aqueous phase.

The aqueous phase containing the REE was shaken with 5 mL of heptane to remove the octanol. Any traces of octanol (or indeed organic material) would result in incomplete evaporation of the sample. Nebulization of organic material has the potential to clog the sample cones through the buildup of a carbon deposit. The time taken to prepare a sample to the evaporation stage was ~90 min.

The sample was collected in a 100 mL PTFE beaker which was later placed in a filtered air chamber on a hot plate, at 40 °C under an infrared lamp. When the sample had evaporated to 5 mL, it was transferred to a Savillex PTFE conical vial for evaporation to near dryness. For normal nebulization, the sample was taken up in 5 mL of 100 ppb Re and Ru in 5% HNO_3 . For flow injection analysis, this volume was reduced to 1.5 mL.

Measurements. Analyses were carried out on the ICPMS operating in scanning mode. The operating conditions of the ICPMS were set to minimize isobaric interference of the light REE oxides onto the heavy REE to ~2% maximum.⁹ The order of analyses of a set of samples was as follows: blank, standards, followed by samples with a blank after every fifth sample. Each sample was analyzed five times with a 5 min wash of a mixture of 0.1% (v/v) orthophosphoric acid and 5% (v/v) HNO_3 in deionized water. This rinse removed any memory effects and possible contamination from previous samples on the subsequent samples.⁹

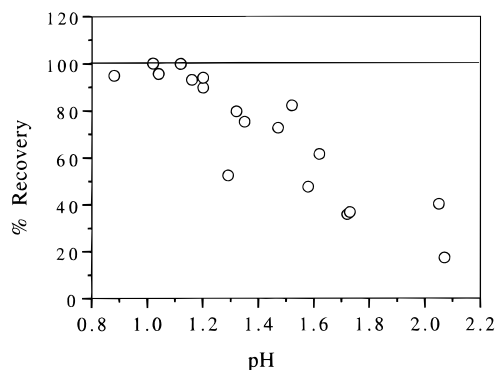


Figure 2. Effect of pH of sample solution on recovery of the REE from the aqueous phase.

RESULTS AND DISCUSSION

As noted earlier, the original method⁶ yields variable recoveries of less than 70% in low ionic strength aqueous samples. In contrast, the salting-out effect in high ionic strength samples (such as seawater) allowed complete recovery of the REE in the pH range of 1.5–4.5. In an attempt to improve the recoveries of the REE in low-salinity waters, the effect of varying the pH of the sample was investigated.

To 1 L samples of sub-boiling distilled water were added 1 mL of a solution containing 100 ppb of each of the REE. These 1 L samples were acidified to pH's varying from 2 to 1 using HCl. The REE were then extracted using the procedure described above. As the pH of the sample was dropped from 2 to 1, there was progressively greater recovery with recoveries reaching greater than 90% at a pH of less than 1.1 (Figure 2).

Separation time of the sample from the organic phase was investigated for its effect in controlling the recovery of the REE. One liter samples of sub-boiling distilled water, spiked with 1 mL of a solution containing 100 ppb of each of the REE were adjusted to a pH of 1.1. The samples were shaken with the diester mix in heptane and left for variable lengths of time before continuing with the extraction process. A separation time of at least 30 min was required to ensure recovery of at least 95%. In the succeeding separation stages it was only necessary to wait until the solution became clear to ensure complete transfer of the REE from one phase to the other.

The speed of separation of the two phases is dependent on the salting-out effect (Shabani, M. B., personal communication, 1993). For higher ionic strength samples the speed of separation is greatly increased. If the ionic strength of the sample is low or unknown, then at least 30 min should be allowed to ensure complete transfer of the REE from the aqueous phase to the organic phase. If the speed of separation needs to be increased to allow for a greater throughput of samples, 10 mL of 14 M HNO_3 and 10 mL of 30% ammonia solution can be added to the sample. The addition of acid and alkali to speed up the rate of separation is not favored due to the risk of contamination.

Detection Limits. Five samples of 1 L of sub-boiling distilled water were passed through the separation process. The processed samples were taken up in 10 mL of 5% HNO_3 containing 100 ppb Re and Ru. With the exception of La, this technique yields detection limits of 0.1–1.6 ppt for the individual REE (Table 2). (The detection limit is defined here as 3 times the standard deviation of five blanks.) The higher La detection limits were a result of a contaminated detector within the instrument at the time

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(9) Kemp, A. J.; Lewis, A. J.; Palmer, M. R., in preparation.

Table 2. ICPMS REE Detection Limits for Normal Nebulization and Flow Injection

	nebulization (ppt)	flow injection (ppt)
La	42	1.7
Ce	1.6	0.19
Pr	0.24	0.07
Nd	0.82	0.25
Sm	0.41	0.48
Eu	0.38	0.15
Gd	0.38	0.34
Tb	0.08	0.05
Dy	0.30	0.27
Ho	0.10	0.06
Er	0.21	0.15
Tm	0.08	0.05
Yb	0.32	0.11
Lu	0.10	0.07

Table 3. Results of Analyses of Samples at 100 ppt Concentrations for All of the REE^a

	sample							
	1	2	3	4	5	6	av	std dev
Ce	100.2	99.8	106.6	105.0	98.8	98.9	101.6	3.4
Pr	100.8	100.2	106.5	106.7	100.6	101.0	102.6	3.1
Nd	99.4	99.7	105.8	105.9	100.0	98.3	101.5	3.4
Sm	100.3	97.5	104.6	104.3	99.3	99.8	101.0	2.9
Eu	100.6	98.8	107.4	105.4	99.7	100.6	102.1	3.5
Gd	101.3	100.0	104.6	104.6	99.3	98.6	101.4	2.6
Tb	101.0	99.8	106.0	105.0	98.5	100.2	101.8	3.0
Dy	100.7	99.0	103.8	104.5	99.7	100.4	101.3	2.3
Ho	99.9	99.6	105.6	103.6	98.0	99.5	101.0	2.9
Er	99.3	99.1	106.8	106.5	99.3	98.7	101.6	3.9
Tm	99.8	98.0	105.1	103.4	98.2	98.4	100.5	3.0
Yb	99.9	99.2	106.7	103.5	99.9	97.8	101.2	3.3
Lu	98.3	99.8	107.7	104.2	100.5	97.5	101.3	3.9
av	100.1	99.3	105.9	104.8	99.4	99.2		
std dev	0.8	0.8	1.2	1.1	0.8	1.1		

^a The data are shown after recalculation using indium internal standard.

of the experiment. Subsequent analyses of blank samples have revealed detection limits for La of the same order as for the other REE.

Precision. Six 1 L samples containing REE at concentrations of 100 ppt were passed through the separation procedure to assess the precision of the method using ¹¹⁵In to monitor the recovery of the REE.⁶ The results of this experiment show that the average concentration of each of the REE is within $\pm 4\%$ of one another at concentrations of 100 ppt (Table 3).

Accuracy. International rock reference materials GSP-1, G2, and AGV-1 were digested⁹ to investigate the accuracy of the separation method on natural materials. The digests were diluted a further 1000 times in sub-boiling distilled water, constituting a 5×10^5 dilution of the original rock sample. These diluted samples were then passed through the extraction process and taken up in 10 mL of 5% HNO₃ containing 100 ppb Re and Ru. The REE concentrations of the diluted samples were recalculated from the actual concentrations measured, and these values are compared to the undiluted samples (before extraction) and the recommended values¹⁰ in Table 4. The analyses show that more than 99.5% of the Ba is removed and that there is complete

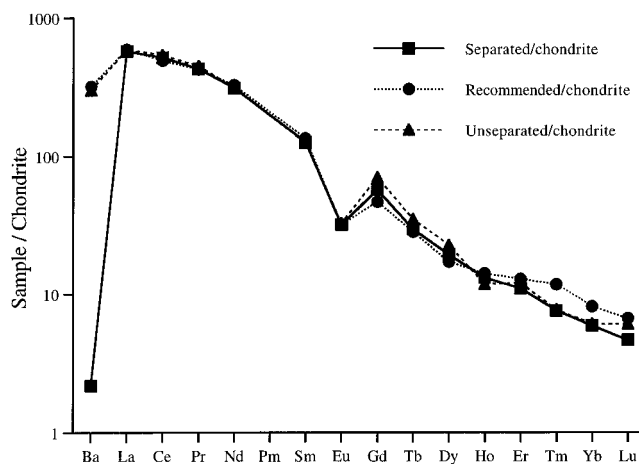


Figure 3. Comparison of chondrite normalized REE patterns of rock reference material GSP-1: (●) recommended values,¹⁰ (■) sample digested, diluted by 5×10^5 and passed through separation procedure, and (▲) sample digested and undiluted. Note that the data have been scaled to take account of the varying dilution factors of the different samples.

recovery of all the REE. The almost complete removal of Ba ensures that there is little isobaric interference from BaO onto the middle REE. Figure 3 shows a comparison of the chondrite normalized REE concentrations of one these rock reference materials (GSP-1) to indicate the suitability of this technique for geological materials. The close agreement between the three data sets demonstrates the accuracy of this technique for samples containing less than 2 ppt of each of the individual REE.

Four samples of GSP-1 were analyzed semiquantitatively (precision $\pm 30\%$) over the mass range 6–240 amu after dilution and REE extraction. These analyses revealed that, apart from the REE, the only other elements recovered during the extraction procedure were Y, Sc, V, Cr, and Zn. The concentrations of Y and Sc in the processed samples were within analytical error of the quoted values, suggesting that the REE extraction method may also be applicable to these elements. However, further work is required to test the accuracy of the Y and Sc data beyond the $\pm 30\%$ precision considered here.

Flow Injection. The level of preconcentration of the extracted sample is limited by the minimum volume required for analysis by ICPMS. In normal nebulization mode, at least 5 mL of sample is required to ensure an acceptable level of precision for the data. Flow injection permits the analysis of small sample volumes and hence allows the sample to be preconcentrated to a greater extent.

There are few techniques that couple the use of flow injection with ICPMS.¹¹ Flow injection ICPMS allows for small-volume throughput using the flow injection, with the sensitivity and speed of ICPMS analysis. Most workers using flow injection have favored this technique because of its high sample throughput and minimal sample consumption. It is the minimal sample consumption that is most desired in ultratrace analysis of REE. It is possible to reduce sample consumption without the use of flow injection by using a reduced uptake rate; however, unless a high-efficiency nebulizer is used, detection limits will be compromised.

The conditions used for flow injection analysis are listed in Table 1. Three analyses were obtained on 1.5 mL of sample, under these conditions. This represents a 3-fold increase in the level of preconcentration of the sample compared to normal nebulization.

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(11) Eaton, A. N.; Hutton, R. C.; Holland, J. G. *Chem. Geol.* **1992**, 95, 63–71.

Table 4. Comparison of REE Concentrations for Three Rock Reference Materials^a

	GSP-1				AGV-1			G2		
	recommended	extraction			recommended	extraction		recommended	extraction	
		before	after	after using FI		before	after		before	after
Ba	1310	1237	8.9		1226	1186	6.0	1882	1713	8
La	184	188	178	183	38	58	58	89	87	92
Ce	399	437	419	414	67	74	70	160	163	164
Pr	52	54	52	52	7.6	8.7	8.3	18	16	17
Nd	196	188	187	181	33	30	31	55	47	51
Sm	26	24	24	26	5.9	5.5	5.7	7.2	6.5	7.2
Eu	2.3	2.4	2.3	1.9	1.64	1.8	1.7	1.4	1.5	1.4
Gd	12.1	18.2	14.8	15.4	5	5.5	5.3	4.3	5.7	5.1
Tb	1.3	1.6	1.4	1.6	0.7	0.64	0.7	0.48	0.51	0.56
Dy	5.5	7.2	6.2	7.5	3.6	3.6	2.3	2.4	2.5	2.4
Ho	1.01	0.85	0.94	0.83	0.67	0.60	0.7	0.4	0.31	0.37
Er	2.7	2.53	2.30	2.7	1.7	1.7	1.9	0.92	0.84	0.91
Tm	0.38	0.25	0.25	0.74	0.34	0.24	0.26	0.18	0.11	0.13
Yb	1.7	1.27	1.23	1.38	1.7	1.5	1.7	0.8	0.50	0.52
Lu	0.21	0.19	0.15	0.016	0.27	0.25	0.23	0.11	0.08	0.08

^a Results shown are the average of five analyses. All concentrations are in ppm and are recalculated after the appropriate dilution has been accounted for. Recommended concentrations are taken from Govindaraju.¹⁰

Table 5. Major Cationic and Anionic Concentrations in Two Geothermal Waters from Iceland^a

	Olkelda	Nesjavellir well 11
Na	600	132
Mg	73	<0.01
K	26	26
Ca	263	0.24
SiO ₂	92	880
CO ₂	1101	74
H ₂ S	<0.01	106
SO ₄ ²⁻	45	4
Cl ⁻	270	90
pH	6.2	8.2

^a All concentrations in ppm.

The detection limits of the flow injection technique were assessed by taking five 1 L aliquots of sub-boiling distilled water and passing them through the extraction process to recover the REE. Once the samples had been evaporated to near dryness, the samples were taken up in 1 mL of 5% HNO₃ containing 100 ppb Re and Ru and analyzed only once by flow injection. With the exception of La (see above), the REE showed detection limits of 0.05–0.5 ppt (Table 2). Comparison of the detection limits of blank samples analyzed by normal nebulization and by flow injection shows an improvement in detection limits of between 15 and 800%, using the flow injection technique.

As a test of the flow injection method, rock reference material GSP-1 was also analyzed using this technique. Five rock digests that had been diluted by a factor of 5×10^5 , relative to the original rock sample, were passed through the extraction procedure and taken up in 1 mL of 5% HNO₃ containing 100 ppb of Re and Ru. Apart from Tm and Lu (which have the lowest abundances in the samples), there is generally good agreement (better than $\pm 10\%$) between the measured and recommended values⁶ (Table 4).

Geothermal Fluids. A preliminary test of the method was made by using the extraction procedure developed here to analyse geothermal fluids collected in Iceland.⁷ These samples were filtered ($<0.45 \mu\text{m}$) on collection and stored in polyethylene bottles with high-purity acids to prevent adsorption of the REE onto the bottle surfaces. The major cation and anion concentrations of

Table 6. REE Concentrations of Geothermal Water from Two Sites in Iceland^a

	Olkelda	Nesjavellir well 11
La	61.9 ± 1.7	30.6 ± 0.4
Ce	69.6 ± 0.7	19.1 ± 0.8
Pr	10.7 ± 1.6	14.4 ± 0.5
Nd	49.3 ± 0.6	7.9 ± 0.2
Sm	14.4 ± 0.9	5.3 ± 0.7
Eu	7.5 ± 0.2	2.5 ± 0.4
Gd	27.8 ± 1.2	13.4 ± 1.1
Tb	4.9	3.3
Dy	38.1 ± 1.4	33.3 ± 3.8
Ho	9.8 ± 0.4	8.8 ± 0.4
Er	27.8 ± 1.9	25.2 ± 1.5
Tm	4.0 ± 0.1	3.0
Yb	25.3 ± 1.7	22.7 ± 0.9
Lu	5.2 ± 0.01	3.9 ± 1.2

^a Details of the sources can be found in Aggarwal.⁷ All concentrations in ppt and errors are 1 σ standard deviation.

these two geothermal fluids are shown in Table 5, indicating the types of matrixes of these fluids. The results of two water analyses are shown in Table 6 and normalized to chondrite in Figure 4.

The concentration of the REE in these two fluids range from 3 to 70 ppt. The standard deviation of these data lies between 3 and 10%, which is acceptable for interpretation of the data. The chondrite normalized patterns are smooth (Figure 3), suggesting that the data are reliable, as irregular REE patterns (with the exception of redox-sensitive elements such as Ce and Eu) are generally indicative of inaccurate data.

CONCLUSIONS

The analytical technique described here is applicable to the determination of REE concentrations in a wide range of aqueous solutions. The method uses a diester mix of 65% bis(2-ethylhexyl) hydrogen phosphate and 35% 2-ethylhexyl dihydrogen phosphate in heptane to allow transfer of the REE into an organic phase. The pH of the solution (<1.1) and the time taken for the first separation stage (≥ 30 min) are crucial to ensure complete recovery of the REE from the aqueous phase.

Matrix interferences as a result of the preconcentration are completely removed. The near complete removal of Ba from the

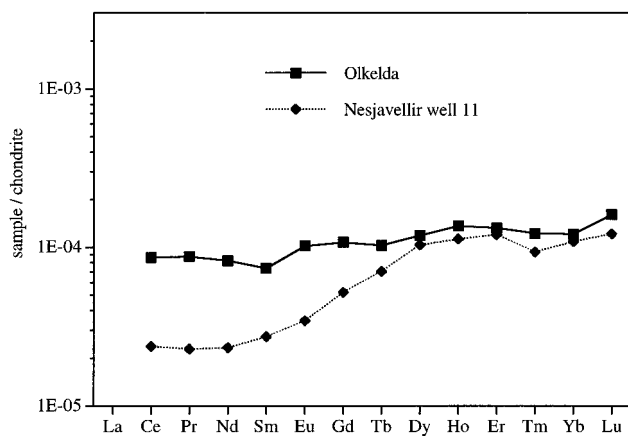


Figure 4. Chondrite normalized REE concentrations for two geothermal fluids collected from Iceland.

sample by the extraction results in no isobaric interference of the BaO onto the middle REE such as Sm, Eu, and Gd.

Detection limits during normal nebulization are less than 0.5 ppt (0.5 pmol kg^{-1}) for most of the REE. Use of flow injection analysis allows the sample to be preconcentrated to smaller volumes and on average results in a 3-fold improvement in detection limits.

There is good agreement between the REE concentrations obtained for rock digests passed through the separation procedure and the recommended values, demonstrating that there is no fractionation of the REE during the extraction procedure. This suggests that the technique is applicable to natural samples where light REE abundances are frequently 2 orders of magnitude greater than those of the heavy REE. This has been tested by analyses of Icelandic geothermal waters, which show smooth REE patterns characteristic of natural samples.

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